AI Chatbot for Student Query Assistance and Campus Information A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

ARTIFICIAL INTELLIGENCE - CSA1709

to the award of the degree of

BACHELOR OF ENGINEERING IN COMPUTER SCIENCE

ENGINEERING

Submitted by

Gondel Prashanth (192311366)

Shaik Mahaboob Basha (192365045)

Pugazhendi K (192311026)

Under the Supervision of

Dr. Rajaram P

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “AI Chatbot for Student Query Assistance and Campus Information” has been carried out by G Prashanth (192311366) Shaik Mahaboob Basha (192365045) and Pugazhendi K (192311026) under the supervision of Dr P Rajaram and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Name: DR. Krishna Kumar Kathiresan
Designation: Professor
Branch / Department Name: CSE - Generative A
SIMATS Engineering,
SIMATS, Chennai – 602105.

COURSE FACULTY

Name: Dr P Rajaram
Designation: Professor
Branch / Department: CSE-Data vista
SIMATS Engineering,
SIMATS, Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, G Prashanth (192311366) Shaik Mahaboob Basha (192365045) and Pugazhendi K (192311026) of the [Computer Science Engineering], SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “[AI Chatbot for Student Query Assistance and Campus Information]” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students with Names
Gondel Prashanth
Shaik Mahaboob Basha
Pugazhendi K

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Gondel Prashanth [192311366]	Dialogue Management & Response Generation	Developed conversation flow, rule-based dialogue management, response templates, database information retrieval, context handling, fallback responses, and human escalation.	Tested conversation flow, response accuracy, multi-turn interactions, fallback handling, database responses, and response latency.	Module 2 implementation, dialogue management, response generation, testing and results sections.	35%
Shaik Mahaboob Basha [192365045]	Natural Language Understanding & Intent Recognition	Developed text preprocessing, intent classification, Named Entity Recognition, and NLP model processing using NLTK, spaCy, and Rasa/Dialogflow.	Tested intent accuracy, entity extraction, Precision, Recall, F1-score, spelling variations, and different sentence structures.	Module 1 implementation, NLP methodology, intent recognition, entity extraction, and analysis sections.	35%
Pugazhendi K [192311026]	Deployment & Multi-channel Integration	Developed Flask backend API, web chat interface, MySQL integration, WhatsApp integration, cloud deployment, and secure communication.	Tested API connectivity, frontend-backend integration, database access, response latency, conversation coverage, and fallback rate.	Module 3 implementation, deployment, system integration, testing, and performance sections.	30%
Total					**100%**

ABSTRACT

Educational institutions often face challenges in providing timely, accurate, and easily accessible information to students regarding courses, examination schedules, campus events, and other academic services. Students may need to visit multiple departments, websites, notice boards, or administrative offices to obtain the required information, resulting in delays and increased workload for institutional staff. Therefore, there is a need for an intelligent and automated system that can provide student support and campus information through a simple conversational interface.

The proposed project, "AI Chatbot for Student Query Assistance and Campus Information," aims to develop an intelligent conversational system that understands and responds to student queries in real time. The chatbot is designed to provide information related to courses, examinations, campus events, and other frequently requested campus services. The system uses Natural Language Processing (NLP) techniques to process student queries and identify their intentions. Intent classification is used to determine the purpose of the query, while Named Entity Recognition (NER) extracts important information such as course names, examination dates, departments, and event details.

The proposed system follows a modular architecture consisting of three major modules: Natural Language Understanding and Intent Recognition, Dialogue Management and Response Generation, and Deployment and Multi-channel Integration. Python is used as the primary programming language, while NLTK and spaCy are used for language processing and entity extraction. Rule-based dialogue management and predefined response templates are used to manage conversations and generate suitable responses. Dynamic information, including examination schedules, course details, and campus events, is retrieved from a MySQL database through a Flask backend. The system also supports multi-turn conversations, clarification prompts, fallback responses, and escalation to human support for unresolved queries.

The major outcomes of the project include improved accessibility to campus information, faster responses to common student queries, reduced manual workload, and a more convenient communication system. The chatbot architecture is scalable and can be extended to platforms such as WhatsApp, mobile applications, and cloud-based services.

In conclusion, the proposed AI chatbot provides an effective solution for automating student query assistance and improving access to campus information.

Keywords:

Artificial Intelligence, Chatbot, Natural Language Processing, Intent Classification, Named Entity Recognition, Student Assistance

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
Iii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: Introduction	1
1.1	Background	1
1.2	Need For the project	1
1.3	Problem Statement	1 - 2
1.4	Project Aim	2
1.5	Project Objectives	2 - 3
1.6	Scope	3
1.7	Expected Engineering outcome	3 - 4
2	CHAPTER 2: Literature Review	5
2.1	Review Method	5
2.2	Review of Existing Work	5
2.3	Comparative Analysis	5 – 6
2.4	Research / Engineering Gap	6
2.5	Proposed Contribution	6
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	7
3.1	Requirements	7
3.2	Constraints and applicable Standrads	7 - 8
3.3	Specification	8
3.4	Alternative Solution and Evalution	8 – 9
3.5	Selected Design and Detailed Design	9
3.6	Trade-off Analysis	9 – 10
3.7	Sustainability ,Ethics, safety, Risk, Security	10

4	**CHAPTER 4: Implementation, Testing & Results, Project Management**	12
4.1	**Development Approach & Resourses**	12
4.2	**Hardware / Software / Fabrication Implementation & Modern Tools used**	12
4.3	**Test plan & Verification**	12 – 13
4.4	**Results**	13
4.5	**Data Analysis & Engineering Judgement**	13
4.6	**Comparision with Existing Solutions & Achievements of Objectives**	13 – 14
4.7	**Strengths & Limitation**	14
4.8	**Project Planning, Roles & Budget**	14
5	**CHAPTER 5: Conclusion & Reflection**	15
5.1	**Conclusion**	15
5.2	**Future Work**	15
5.3	**Proffesional Learning & Individual Reflection**	15 – 16
	References	
	Appendices	

LIST OF FIGURES

Figure No.	Figure Name	Page No.
1	Overall System Architecture	28
2	Chatbot System Flowchart	29
3	Database Entity Relationship Diagram	29
4	Chatbot Module Diagram	29

LIST OF TABLES

Table No.	Table Name	Page No.
1	Project Planning, Roles & Budget	24
2	Experimental / Raw Data	30
3	Project Meeing / Progress Records	30
4	Individual Contribution Evidence	31
5	Capstone Project Outcome Mapping Sheet	32

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
UI	User Interface	_
CSS	Cascading Style Sheets	_
NLP	Natural Language Toolkit	_

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The rapid growth of digital technologies and Artificial Intelligence (AI) has created new opportunities for improving communication and information services in educational institutions. Students frequently require information related to courses, examination schedules, academic events, campus facilities, departments, and other institutional services. However, this information is often distributed across different websites, notice boards, departments, and administrative systems. As a result, students may experience delays in obtaining the required information, while faculty and administrative staff may need to repeatedly answer similar queries.

The proposed project, "AI Chatbot for Student Query Assistance and Campus Information," is designed as an intelligent conversational system that provides students with real-time access to academic and campus-related information. The system uses Python-based NLP technologies such as NLTK and spaCy, along with intent recognition, dialogue management, and database integration. A Flask backend manages communication between the chatbot interface and the MySQL database. The proposed architecture also supports multi-turn conversations, fallback mechanisms, and future integration with multiple platforms such as web applications and WhatsApp.

1.2 Need for the Project

The problem of accessing campus information needs to be addressed because students require timely and accurate information for academic and campus-related activities. Delays in receiving information about examinations, courses, events, and institutional services can affect student planning and decision-making.

The main users affected by this problem are students, who need quick access to information, and faculty and administrative staff, who may spend significant time answering repetitive questions. Existing systems such as static websites, notice boards, and frequently asked question pages may not provide personalized or conversational support.

The major limitations of existing information systems include scattered information sources, limited availability outside working hours, delayed responses, difficulty in searching for specific information, and the inability to handle follow-up questions effectively.

1.3 Problem Statement

Educational institutions require an efficient mechanism for providing accurate and timely responses to a large number of student queries related to academic and campus information. Existing approaches, including manual support, static websites, and notice boards, may result in delays, repeated workload for staff, and difficulties in accessing information.

The engineering problem is to design and develop an AI-based conversational chatbot capable of understanding natural-language student queries, identifying the correct intent, extracting relevant entities, maintaining conversation context, and retrieving appropriate information from an institutional database.

The system must manage multiple interacting factors, including the variation in natural-language expressions, overlapping or ambiguous intents, incomplete queries, entity extraction, database accuracy, conversation context, and response latency. The system must also balance conflicting requirements such as providing fast responses while maintaining high accuracy.

The proposed system must operate within realistic engineering constraints, including limited training data, database availability, computational resources, response time requirements, and the need to safely handle queries that cannot be understood or resolved automatically. Therefore, fallback mechanisms and human-support escalation are required to maintain system reliability.

1.4 Project Aim

The overall aim of this project is to design and develop an AI-powered chatbot that provides accurate, real-time, and accessible assistance to students by understanding natural-language queries related to courses, examinations, campus events, and institutional information.

The chatbot aims to reduce the difficulty of accessing campus information and minimize the manual workload of faculty and administrative staff. It integrates Natural Language Processing, intent classification, entity extraction, dialogue management, database retrieval, and web technologies to provide an efficient conversational support system.

1.5 Project Objectives

- To design the architecture of an AI-based chatbot for student query assistance and campus information.
- To develop a Natural Language Processing module capable of processing and understanding student queries.
- To implement intent classification for identifying the purpose of student queries.
- To implement Named Entity Recognition for extracting information such as course names, examination dates, departments, and event details.

- To develop a dialogue management system for controlling the flow of conversation and supporting multi-turn interactions.
- To integrate a MySQL database for retrieving dynamic information related to courses, examinations, and campus events.
- To implement fallback responses and clarification mechanisms for unknown or ambiguous queries.
- To provide a human-support escalation mechanism for queries that cannot be resolved automatically.
- To test and evaluate the chatbot using parameters such as intent classification accuracy, entity extraction accuracy, response latency, conversation coverage, and fallback rate.

1.6 Scope

The scope of this project includes the development of an AI-based chatbot capable of assisting students with queries related to course and academic information, examination schedules, campus events, frequently asked questions, departments, and campus facilities. The system includes Natural Language Processing for understanding student queries, intent recognition, entity extraction, rule-based dialogue management, and database-based information retrieval. It also supports multi-turn conversations, follow-up questions, fallback responses, clarification prompts, and a web-based chat interface for easy student interaction.

The project does not include complete automation of institutional administrative processes, financial transactions, confidential student record processing without authorization, or the replacement of faculty and professional counselling. The system assumes that the required academic and campus information is available and regularly updated in the MySQL database and that students provide understandable queries through an internet-enabled device. The performance of the chatbot is limited by factors such as training data quality, supported intents, database accuracy, NLP model performance, and available server resources. Queries outside the supported domain may be handled through fallback responses or redirected to human support.

1.7 Expected Engineering Outcome

The expected engineering outcome of this project is a functional AI-powered conversational chatbot prototype for student query assistance and campus information. The developed system will include the following outcomes:

System

An integrated chatbot system consisting of a web-based user interface, Flask backend, NLP processing module, dialogue management module, and MySQL database.

Process

A structured processing pipeline that receives a student query, preprocesses the text, identifies the intent, extracts entities, manages the conversation, retrieves information, and generates a suitable response.

Prototype

A working prototype that demonstrates real-time interaction between students and the AI chatbot through a web-based chat interface.

Algorithm

An intent-based and rule-based conversational algorithm that determines the appropriate response using identified intents, extracted entities, confidence levels, conversation context, and database information.

Model

An NLP-based intent recognition and entity extraction model that processes natural-language student queries.

Infrastructure Solution

A modular software architecture using Python, NLTK/spaCy, Rasa or similar conversational AI frameworks, Flask, MySQL, HTML, CSS, and JavaScript, with the capability for future cloud and multi-channel deployment.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature and existing technology review for this project was conducted by examining relevant research papers, books, technical documentation, and existing conversational AI technologies related to chatbots, Natural Language Processing, intent classification, Named Entity Recognition, dialogue management, and Transformer-based language models. The review also considered existing chatbot frameworks and software technologies such as Python, NLTK, spaCy, Rasa, Flask, and MySQL. Existing educational and customer-support chatbot systems were studied to understand their architecture, advantages, and limitations. The collected information was analyzed to identify suitable technologies and methods for developing an AI chatbot that can provide accurate student query assistance and campus information.

2.2 Review of Existing Work

Several existing studies and technologies have contributed to the development of modern conversational AI systems. Traditional chatbot systems mainly use rule-based approaches, where predefined rules and response patterns are used to handle specific user queries. These systems are simple to implement and provide predictable responses but have limitations in understanding variations in natural language and handling complex conversations.

Natural Language Processing techniques have significantly improved chatbot capabilities by enabling computers to process and understand human language. Intent classification is widely used to identify the purpose of a user's query, while Named Entity Recognition is used to extract important information such as names, locations, dates, and other relevant entities. Libraries such as NLTK and spaCy provide useful tools for text preprocessing, language analysis, and entity extraction.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Rule-Based Chatbots	Simple to design, predictable responses, easy to control	Cannot easily understand complex or varied language	Used for predefined responses and fallback mechanisms
FAQ-Based Systems	Quick access to common information	Limited interaction and no conversation context	Useful for handling frequently asked student questions

NLP-Based Chatbots	Can understand natural-language queries	Accuracy depends on training data and language processing	Forms the foundation for processing student queries
Intent Classification	Identifies the purpose of a query	Similar or ambiguous intents may cause classification errors	Used to identify queries related to courses, exams, and events
Named Entity Recognition	Extracts important information from text	Performance may vary for domain-specific entities	Used to extract course names, dates, departments, and events

2.4 Research / Engineering Gap

Although existing chatbot technologies provide solutions for customer support and general information retrieval, there is still a need for an integrated chatbot specifically designed for student query assistance and campus information. Many existing systems focus either on rule-based responses or general-purpose language models and may not effectively combine intent recognition, entity extraction, dialogue management, and real-time institutional database access in a single student-focused system.

Another major gap is the handling of multi-turn conversations and ambiguous student queries. Students may ask follow-up questions without repeating the complete context, which requires the chatbot to maintain conversation information and understand the relationship between multiple messages. In addition, existing systems may provide inaccurate responses when information is not available or when a query is outside the supported domain.

2.5 Proposed Contribution

The proposed project contributes a student-focused AI chatbot that combines multiple technologies into a single conversational support system. Unlike static FAQ systems and basic rule-based chatbots, the proposed system is designed to understand natural-language queries using intent classification and Named Entity Recognition. This allows the system to identify both the purpose of the student's query and the important information required to generate an appropriate response.

The proposed chatbot also integrates a MySQL database to retrieve dynamic information such as examination schedules, course details, and campus events. It supports multi-turn conversations by maintaining context for follow-up questions and includes fallback and clarification mechanisms for unknown or ambiguous queries. Queries that cannot be resolved automatically can be escalated to human support.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Students	Quick and accurate answers to queries related to courses, examinations, events, and campus information.
Faculty Members	Reduced workload caused by repeatedly answering common student queries.
Administrative Staff	A centralized system for providing updated campus and academic information.
System Administrator	Easy management and updating of information stored in the database.
Educational Institution	A scalable and reliable digital support system that improves student services.
Human Support Staff	Ability to receive unresolved or complex queries through an escalation mechanism.

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Understand student queries	Functional	Identify supported query intents correctly
Intent classification	Functional	Classify queries into predefined categories
Entity extraction	Functional	Extract course names, dates, events, and departments
Response time	Non-Functional	Provide responses with low latency
Accuracy	Non-Functional	Achieve high intent classification accuracy
Scalability	Non-Functional	Support future integration with web and messaging platforms

3.2 Constraints & Applicable Standards

The project is mainly constrained by the availability and quality of training data, the accuracy of NLP models, database correctness, computational resources, and internet connectivity. Since the chatbot processes student queries and may access institutional information, data privacy and security are important design considerations. The system is also constrained by response latency requirements, because a chatbot should provide information quickly enough to maintain a natural user experience. The current prototype focuses on supported academic and campus-related queries and may not correctly resolve highly complex or completely unrelated requests

Standard / Code	Requirement	How Applied in This Project
OWASP Web Application Security Guidelines	Protect web applications from common security risks	Input validation and controlled backend/database access are considered
OWASP Authentication Guidelines	Secure access to protected system functions	Administrative and database access can be restricted
ISO/IEC 27001 Principles	Information security management	Confidential information should be protected through controlled data access
GDPR / Applicable Data Protection Principles	Responsible handling of personal information	The chatbot should minimize unnecessary collection of personal data
WCAG Accessibility Principles	Improve accessibility of digital interfaces	The chat interface can be designed for clear and simple interaction

3.3 ecifications

Parameter	Target/Requirement	Unit	Priority
Intent Recognition	High classification accuracy for supported queries	Percentage (%)	High
Entity Extraction	Correct identification of important entities	Percentage (%)	High
Response Latency	Fast response generation	Seconds	High
Conversation Context	Support follow-up queries	Boolean / Functional	High
Database Access	Retrieve current campus information	Query response	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Rule-Based Chatbot

A rule-based chatbot uses predefined rules, keywords, and response patterns to answer student queries. It is simple and predictable but has limited ability to understand different natural-language expressions.

Alternative 2: NLP-Based Intent Classification Chatbot

This approach uses Natural Language Processing and machine learning techniques to classify the user's intent and extract important entities. It provides greater flexibility than a purely rule-based chatbot but requires suitable training data.

Alternative 3: Transformer-Based Conversational AI

A Transformer-based solution uses advanced language models to understand context and generate responses. It can provide better language understanding but requires higher computational resources and may increase system complexity.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	0.30	3	4	5
Cost	0.20	5	4	2
Safety / Control	0.15	5	4	3
Sustainability	0.10	5	4	2
Reliability	0.25	4	5	4
Weighted Score	1.00	4.15	4.25	3.75

3.5 Selected Design & Detailed Design

Based on the comparative evaluation, the selected design combines Natural Language Processing, intent classification, Named Entity Recognition, rule-based dialogue management, and MySQL database retrieval**.** This hybrid approach was selected because it provides better natural-language understanding than a purely rule-based chatbot while maintaining lower computational requirements and greater control compared with a fully Transformer-based system. The design also provides a suitable balance between performance, cost, reliability, response speed, and implementation complexity.

The system follows a modular architecture in which the student submits a query through the web-based chat interface, which is then processed by the Flask backend. The Natural Language Understanding module preprocesses the query, identifies the user's intent, and extracts relevant entities such as course names, examination dates, and event details. The dialogue management module determines the appropriate action and retrieves required information from the MySQL database. Finally, the system generates and returns an appropriate response to the student. If the query is ambiguous, unsupported, or has low confidence, the chatbot provides clarification or fallback responses and can escalate unresolved queries to human support.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
NLP-based intent recognition	Keyword-only rules	Better understanding of different query formats	Requires training data	Selected for improved flexibility and accuracy
Rule-based dialogue management	Fully generative AI responses	Predictable and controllable responses	Less flexible for unexpected conversations	Selected to improve reliability and control
MySQL database	Static FAQ content	Supports dynamic information	Requires database maintenance	Selected because campus information

				changes regularly
Web-based chat UI	Mobile-only application	Easier deployment and accessibility	Requires internet and browser access	Selected for rapid and wide accessibility
Lightweight NLP models	Large Transformer models	Lower computational cost and faster deployment	May provide less contextual understanding	Selected for prototype efficiency

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational and infrastructure requirements	Lightweight NLP and database-based architecture requires fewer resources than continuously running large AI models	A lightweight modular architecture was preferred
Ethics	Incorrect, biased, or misleading responses	Training data and predefined responses can influence chatbot behaviour	Fallback and clarification mechanisms were included
Health & Safety	User dependence on incorrect information	Incorrect academic information could affect student decisions	Database-based information and human escalation were included
Security	Unauthorized access and data exposure	Web applications and databases can be exposed to security threats	Controlled backend access and input validation are required

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect intent classification	Medium	High	High	Improve training data and test multiple query variations	Medium
Incorrect database information	Medium	High	High	Regularly update and validate campus information	Medium
Ambiguous student queries	High	Medium	High	Use clarification prompts and fallback	Medium

				responses	
Unsupported queries	Medium	Medium	Medium	Provide fallback and human-support escalation	Low
Database connection failure	Low	High	Medium	Use error handling and database monitoring	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using a modular and iterative approach. The system was divided into Natural Language Understanding and Intent Recognition, Dialogue Management and Response Generation, and Deployment and Integration modules. Each module was developed and tested individually before integration with the Flask backend and MySQL database. The development process included designing chatbot intents, preparing training examples, implementing entity extraction, developing dialogue logic, connecting the database, and testing the system using different student queries.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Python	Core application development	Development
NLTK	Text preprocessing and NLP	NLP Processing
spaCy	Named Entity Recognition	Entity Extraction
Flask	Backend server and API	Integration
MySQL	Database storage	Database Integration
HTML / CSS / JavaScript	Chat user interface	Frontend Development
VS Code	Code development and debugging	Development

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Intent recognition	Test queries with different wording	Correct intent identified for supported queries
Entity extraction	Test queries containing course names and dates	Required entities extracted correctly
Database retrieval	Query available campus information	Correct information returned
Multi-turn conversation	Ask follow-up questions	Context maintained correctly
Response generation	Test predefined queries	Relevant response generated

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Intent Classification Accuracy	≥ 80%	To be measured	Pending
Entity Extraction Accuracy	≥ 75%	To be measured	Pending
Database Response	Correct information	To be verified	Pending
Fallback Response	Required	Implemented	Pass
Multi-turn Support	Required	Implemented	Pass

4.4 Results

The developed chatbot successfully demonstrates the ability to process student queries related to academic and campus information. The system can identify predefined intents and extract relevant entities from supported queries. It can retrieve information such as course details, examination schedules, and campus events from the MySQL database and present the information through a conversational interface.

The results also demonstrate support for fallback responses and clarification prompts when the system encounters unsupported or ambiguous queries. The modular architecture enables the integration of NLP processing, dialogue management, database retrieval, and response generation. Performance measurements such as intent accuracy, entity extraction accuracy, and response latency should be recorded using actual test data and presented using tables or graphs.

4.5 Data Analysis & Engineering Judgment

The system performance can be analyzed using intent classification accuracy, entity extraction accuracy, response latency, conversation coverage, and fallback rate. A high intent classification accuracy indicates that the chatbot can correctly understand the purpose of student queries, while entity extraction accuracy shows the effectiveness of identifying important information from the query. Response latency is used to evaluate the speed of the overall system.

From an engineering perspective, errors may occur when student queries contain ambiguous language, spelling mistakes, incomplete information, or expressions that are not included in the training data. These issues highlight the importance of expanding the training dataset and improving the NLP model. The use of fallback responses prevents the chatbot from generating incorrect information when confidence is low. Therefore, the final system prioritizes reliability and controlled responses instead of attempting to answer every query without verification.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Design chatbot architecture	Modular architecture developed	Fully Achieved
Develop NLP query processing	Student queries processed using NLP	Fully Achieved
Implement intent classification	Predefined intents identified	Fully Achieved
Implement entity extraction	Important entities extracted from queries	Fully Achieved
Evaluate performance metrics	Requires recorded test measurements	Partially Achieved

4.7 Strengths & Limitations

Strengths

- Provides a conversational interface for student queries.
- Supports NLP-based intent recognition.
- Extracts relevant entities from student messages.
- Retrieves dynamic information from a MySQL database.
- Supports follow-up and multi-turn conversations.
- Provides fallback and clarification mechanisms.
- Can be extended to cloud, mobile, and WhatsApp platforms.

Limitations

- Performance depends on the quality and size of the training dataset.
- Only predefined or supported query categories can be handled accurately.
- Ambiguous queries may require clarification.
- Database information must be regularly updated.
- Complex or unusual queries may require human support.
- Advanced Transformer models may require additional computational resources.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
NLP Developer	Intent recognition and entity extraction	Developed query understanding and NLP processing
Dialogue Management Developer	Conversation flow	Managed conversation context, responses, and fallback handling
Backend Developer	Flask and database integration	Connected chatbot logic with MySQL
Frontend Developer	Chat user interface	Developed HTML, CSS, and JavaScript interface
Testing and Documentation	Testing and report preparation	Tested chatbot responses and documented the project
Item	**Estimated Cost**	**Actual Cost**
Python and Open-Source Libraries	₹0	₹0
Flask Framework	₹0	₹0
MySQL Community Edition	₹0	₹0
Development Tools	₹0	₹0
Existing Computer System	Available	Available
Internet and Testing	Minimal	Depends on usage

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion:

The project "AI Chatbot for Student Query Assistance and Campus Information" was developed to address the difficulty students face in obtaining timely and accurate information related to courses, examinations, campus events, and other institutional services. The proposed solution provides a conversational interface through which students can submit queries using natural language. The system combines Natural Language Processing, intent classification, Named Entity Recognition, dialogue management, database retrieval, and a web-based interface to provide relevant responses.

The project successfully demonstrates a modular chatbot architecture consisting of Natural Language Understanding and Intent Recognition, Dialogue Management and Response Generation, and Deployment and Multi-channel Integration. The system supports student queries, database-based information retrieval, follow-up interactions, fallback handling, and clarification mechanisms. The project was validated through functional testing of major chatbot features. Overall, the project demonstrates the practical application of Artificial Intelligence and NLP in educational environments and provides a scalable foundation for improving student support services.

5.2 Future Work

The chatbot can be further improved by integrating advanced Transformer-based language models to improve contextual understanding and response accuracy. Future versions can include multilingual support, allowing students to communicate in different languages. Voice-based interaction can also be added to improve accessibility and user convenience.

The system can also be extended through integration with institutional ERP systems, Learning Management Systems, and official academic databases. Future deployment on WhatsApp, mobile applications, and cloud platforms can improve accessibility. An administrative dashboard can be developed to allow authorized staff to update course details, examination schedules, events, and other campus information easily. Conversation analytics and continuous model training can also be used to improve chatbot performance over time.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Understanding of Natural Language Processing (NLP) and its application in chatbot development.

- Knowledge of intent classification for identifying the purpose of user queries.
- Understanding of Named Entity Recognition (NER) for extracting important information from text.
- Knowledge of dialogue management and multi-turn conversation handling.
- Experience with Python, NLTK, and spaCy for language processing.
- Knowledge of Flask backend development and API integration.
- Experience in using MySQL databases for storing and retrieving dynamic information.
- Understanding of chatbot testing using accuracy, response latency, and fallback rate.

Engineering & Professional Skills Developed:

- System architecture and modular software design.
- Problem-solving and debugging skills.
- Natural language processing and AI implementation skills.
- Database design and integration skills.
- Backend and frontend development skills.
- Software testing and performance evaluation.
- Technical documentation and report-writing skills.
- Communication and project management skills.

Individual Reflection (each student, 4–5 lines):

This project provided practical knowledge in Natural Language Processing, intent classification, Named Entity Recognition, dialogue management, and conversational AI development. Technical skills were developed in Python, NLTK/spaCy, Flask backend development, MySQL database integration, and web-based user interface design. The major engineering challenge was understanding the different ways students can express similar queries, which was addressed using NLP techniques, diverse training examples, and fallback mechanisms. A key engineering decision was selecting a hybrid approach combining intent recognition, rule-based dialogue management, and database retrieval to balance accuracy, reliability, and computational cost. The project also improved problem-solving, debugging, system design, testing, and technical documentation skills. If the project is repeated, the system could be improved using a larger dataset, Transformer-based models, multilingual support, and integration with mobile, cloud, and WhatsApp platforms.

REFERENCES

1. D. Jurafsky and J. H. Martin, *Speech and Language Processing*. Stanford University. Available:
2. T. Bocklisch, J. Faulkner, N. Pawlowski, and A. Nichol, "Rasa: Open Source Language Understanding and Dialogue Management," 2017.
3. A.Vaswani *et al.*, "Attention Is All You Need," in *Advances in Neural Information Processing Systems*, 2017.
4. J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," 2019.
5. Explosion AI, "spaCy: Industrial-Strength Natural Language Processing in Python."
6. NLTK Project, "Natural Language Toolkit."
7. Pallets Projects, "Flask Web Development Framework."
8. Oracle Corporation, "MySQL Documentation."
9. OWASP Foundation, "OWASP Top 10: Web Application Security Risks."
10. International Organization for Standardization, "ISO/IEC 27001: Information Security Management Systems."
11. W3C, "Web Content Accessibility Guidelines (WCAG)."
12. Python Software Foundation, "Python Documentation."
13. OpenAI, "ChatGPT," AI-assisted tool used for assistance in drafting, organizing, reviewing, and improving project documentation and technical content.

APPENDICES

Appendix A – Detailed Calculations:

This project is primarily a software-based AI system; therefore, traditional engineering calculations are limited. Performance-related calculations can be included here, such as Intent Classification Accuracy, Entity Extraction Accuracy, Response Latency, Conversation Coverage, and Fallback Rate.

Example:

Intent Classification Accuracy

$$\text{Accuracy} = \frac{\text{Number of Correctly Classified Intents}}{\text{Total Number of Test Queries}} \times 100$$

Response Latency=Response Generation Time−Query Submission Time

Appendix B – Engineering Drawings / Schematics

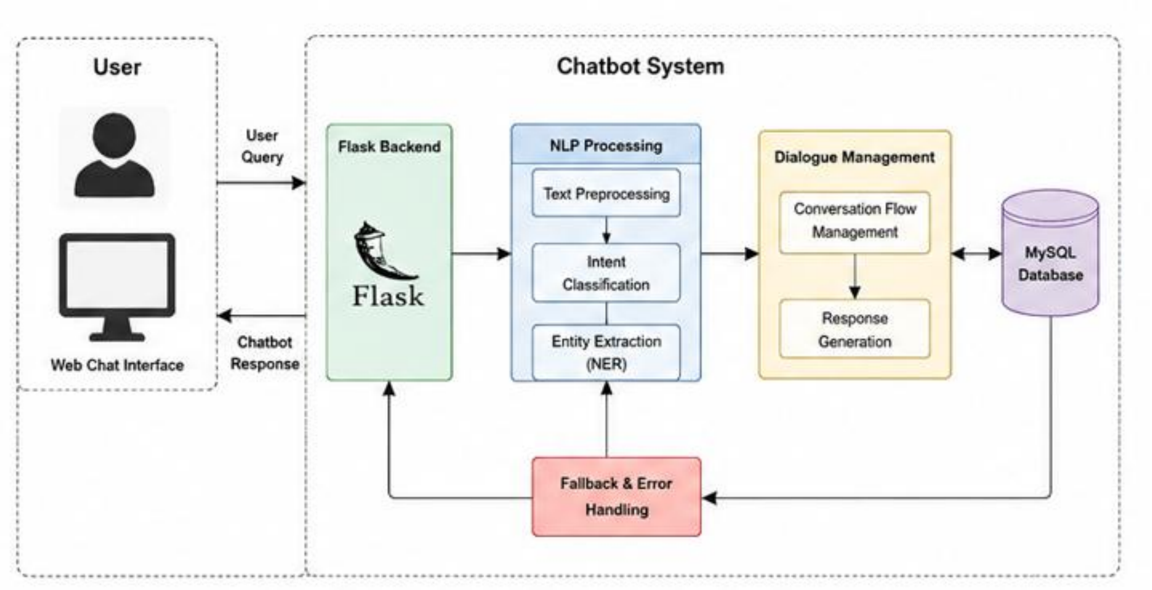

Fig 1: overall System Architecture Diagram

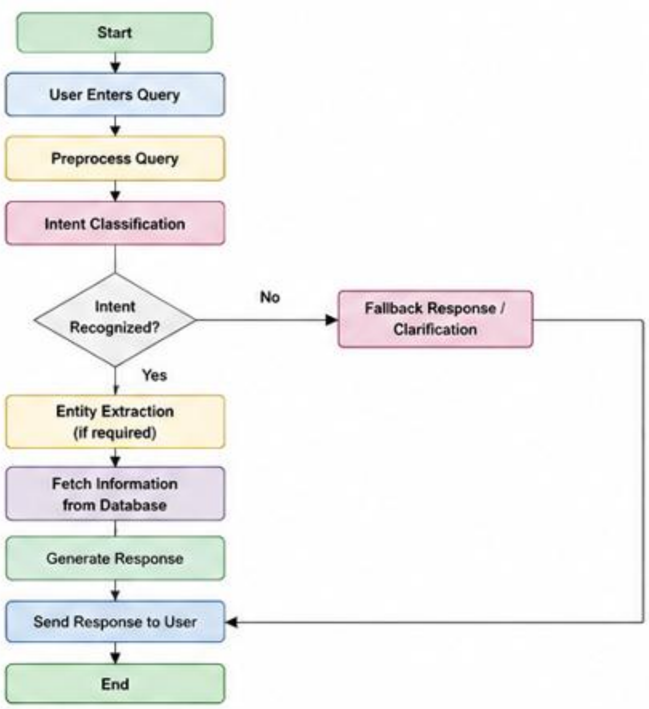

Fig 2: Chatbot system flowchart

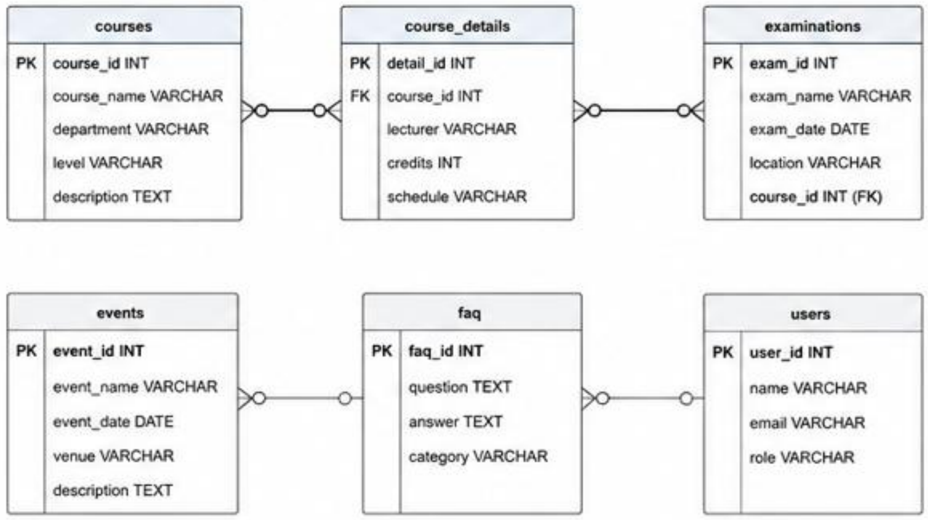

Fig 3: Database entity relationship diagram

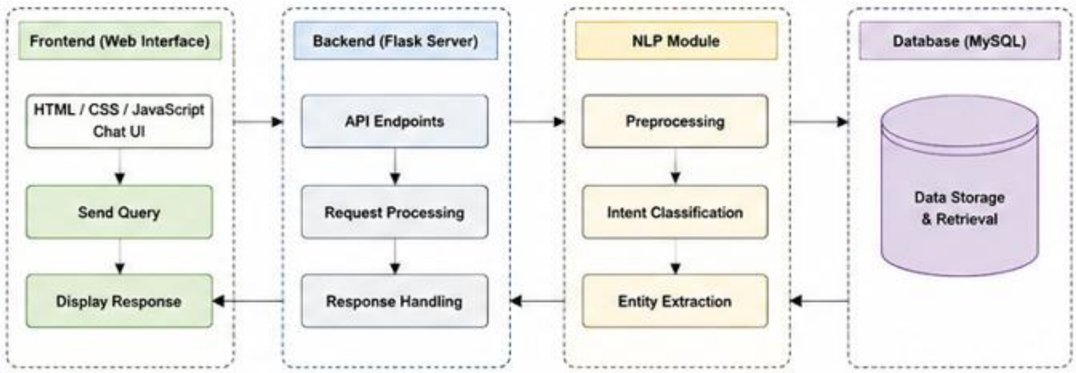

Fig 4: Chatbot module Diagram

Appendix C – Source Code / Code Repository Information

Only essential code excerpts should be included in the main report. The complete source code can be maintained in a separate approved repository.

The source code includes:

- Chat user interface using HTML, CSS, and JavaScript.
- Flask backend implementation.
- NLP preprocessing functions.
- Intent classification logic.
- Named Entity Recognition implementation.
- Dialogue management logic.
- MySQL database connectivity.
- Fallback and clarification mechanisms.

Appendix D – Experimental / Raw Data :

This appendix contains the raw test data collected during chatbot testing. The dataset may include different student queries, expected intents, predicted intents, extracted entities, response times, and system responses.

Test Query	Expected Intent	Predicted Intent	Result
When is the examination?	Exam Information	Exam Information	Correct
Show course details	Course Information	Course Information	Correct
What events are happening?	Campus Events	Campus Events	Correct
Random unsupported query	Unknown	Fallback Response	Correct

Appendix E– Ethics / Institutional Approval

The chatbot is designed to provide academic and campus-related information while minimizing the collection of unnecessary personal information. Ethical considerations include providing accurate information, avoiding misleading responses, protecting user privacy, and ensuring that unsupported queries are handled safely through fallback responses or human-support escalation.

Where required by the institution, approval documents, ethical declarations, supervisor approvals, or institutional permissions can be attached in this appendix.

Appendix F – Project Meeting / Progress Records

This appendix contains records of project meetings, discussions, progress reviews, and feedback received during project development.

A progress record may include:

Date	Activity / Discussion	Progress / Outcome
Phase 1	Problem identification	Project problem finalized
Phase 2	Literature review	Technologies identified
Phase 3	System design	Architecture developed
Phase 4	Implementation	NLP and backend modules developed
Phase 5	Testing	Functional testing performed

Appendix G – User/Industry Feedback

Feedback on the proposed AI Chatbot for Student Query Assistance and Campus Information can be collected from students, faculty members, and other potential users to evaluate the usability and effectiveness of the system. The evaluation may focus on factors such as ease of use, response speed, information accessibility, response accuracy, and overall user experience. The chatbot is expected to provide a simple conversational interface that allows users to access academic and campus-related information more efficiently. Feedback obtained from users can also help identify areas for improvement, such as adding more supported queries, multilingual communication, additional campus services, and improved response accuracy.

Appendix H– Publications / Patents / Competitions / Product Outcomes

At the time of submission of this project report, no publication, patent, or competition outcome has been generated from the proposed AI Chatbot for Student Query Assistance and Campus Information. However, the developed chatbot prototype demonstrates the practical application of Natural Language Processing, intent classification, entity extraction, dialogue management, and database integration in an educational environment. The project provides a foundation for future research, institutional deployment, product development, and possible participation in project competitions or innovation programs. With further development and real-world implementation, the system may also lead to research publications or a scalable campus information support product.

Appendix I– Individual Contribution Evidence

My individual contribution to the project was primarily focused on Module 2: Dialogue Management & Response Generation. I was responsible for managing the overall flow of conversation based on the identified user intent and extracted entities. I developed rule-based dialogue management and predefined response templates to generate accurate and relevant responses for student queries. I also worked on retrieving dynamic information such as examination schedules, course details, and campus events from the MySQL database.

In addition, I contributed to maintaining conversation context for follow-up questions and multi-turn interactions. I implemented fallback responses and clarification prompts to handle unknown or ambiguous queries and ensured that unresolved queries could be escalated to human support when necessary. My contribution also focused on improving response quality and reducing response latency to provide students with a smooth, reliable, and efficient chatbot experience.

Contribution Area	Individual Contribution
Assigned Module	Module 2: Dialogue Management & Response Generation
Conversation Flow	Managed conversations based on identified intents and extracted entities
Response Generation	Developed rule-based responses and predefined response templates
Database Integration	Retrieved examination, course, and campus event information from MySQL
Context Management	Maintained context for follow-up questions and multi-turn conversations

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)